SLIP 1

Q1. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics. (13 Marks)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Mall_Customers.csv")

age = df["Age"]

plt.figure(figsize=(6,4))
plt.hist(age, bins=10, edgecolor="black")
plt.title("Histogram of Age")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(5,4))
plt.boxplot(age)
plt.title("Box Plot of Age")
plt.ylabel("Age")
plt.show()

print("Minimum Age:", age.min())
print("Maximum Age:", age.max())
print("Average Age:", age.mean())
print("Median Age:", age.median())
```

Q2. Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed. (12 Marks)

```
import random
import pandas as pd

population = [
    "Aarav", "Riya", "Rahul", "Sneha", "Aditya",
    "Priya", "Rohan", "Neha", "Karan", "Pooja",
    "Vivek", "Anjali", "Aman", "Simran", "Raj"
]

sample_size = 10
sample = random.sample(population, sample_size)
data = []

for person in sample:
    print("\nParticipant:", person)
    age = int(input("Enter age: "))
    gender = input("Enter gender: ")
    satisfaction = int(input("Rate satisfaction (1-5): "))

    data.append({
        "Name": person,
        "Age": age,
        "Gender": gender,
        "Satisfaction": satisfaction
    })

df = pd.DataFrame(data)

print(df)
df.to_csv("survey_data.csv", index=False)
```

```
print("Survey completed successfully!")
print("Data saved as survey_data.csv")
```

SLIP 2

Q1. Use a dataset with multiple variables and perform correlation analysis to determine the strength and direction of relationships between pairs of variables. (13 Marks)

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("Iris.csv")

num_data = data.select_dtypes(include="number")
corr = num_data.corr()

print("Correlation Matrix")
print(corr)

plt.imshow(corr, cmap="coolwarm")
plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Matrix")
plt.show()
```

Q2. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests. (12 Marks)

```
import numpy as np
from scipy import stats

sales = [520, 540, 510, 530, 550,
         525, 560, 535, 545, 555]

population_mean = 500

t_stat, p_value = stats.ttest_1samp(
    sales, population_mean
)

p_value = p_value / 2

print("T-statistic:", round(t_stat, 3))
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Reject Null Hypothesis")
    print("New marketing campaign increased sales.")
else:
    print("Fail to Reject Null Hypothesis")
    print("No significant increase in sales.")
```

SLIP 3

Q1. Utilize a business intelligence tool (e.g., Tableau, Power BI) to extract insights from a dataset and create interactive visualizations for effective data analysis. (13 Marks)

1. Open Tableau Public/Desktop.
2. Click Connect → Microsoft Excel.
3. Select the Superstore Sales dataset.
4. Click Open.
5. Check the fields: Order Date, Sales, Profit, Category and Region.
6. Click Sheet 1.
7. Drag Category to Columns.
8. Drag Sales to Rows.
9. Tableau creates a bar chart.
10. Drag Region to Color.
11. Drag Profit to Tooltip.
12. Create additional charts using Order Date and Sales.
13. Add the worksheets to a Dashboard.
14. Add filters to make the dashboard interactive.
15. Save the workbook.

Q2. Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters. (12 Marks)

```
import numpy as np
import matplotlib.pyplot as plt

population = np.random.randint(1, 101, 1000)

means = []

for i in range(500):
    sample = np.random.choice(population, 30)
    means.append(np.mean(sample))

print("Population Mean:", np.mean(population))
print("Sample Mean:", np.mean(means))
print("Standard Error:", np.std(means))

plt.hist(means, bins=20, edgecolor="black")
plt.title("Central Limit Theorem")
plt.xlabel("Sample Means")
plt.ylabel("Frequency")
plt.show()
```

SLIP 4

Q1. Use a dataset with multiple variables and perform correlation analysis to determine the strength and direction of relationships between pairs of variables. (13 Marks)

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("Iris.csv")

num_data = data.select_dtypes(include="number")
corr = num_data.corr()

print("Correlation Matrix")
print(corr)

plt.imshow(corr, cmap="coolwarm")
plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Matrix")
plt.show()
```

Q2. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions. (12 Marks)

```
import numpy as np
from scipy.stats import chi2_contingency

A = [120, 80]
B = [100, 100]

rate_A = A[1] / sum(A)
rate_B = B[1] / sum(B)

print("Conversion Rate A:", rate_A)
print("Conversion Rate B:", rate_B)

table = [A, B]

chi2, p_value, dof, expected = chi2_contingency(table)

print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Significant difference found.")
    if rate_B > rate_A:
        print("Strategy B is more effective.")
    else:
        print("Strategy A is more effective.")
else:
    print("No significant difference between A and B.")
```